VC_ARG_04 · Operation specification v4.0

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Project: Vibe Coding Industrial, Digitalización Férriz. Process specification of the 100 L batch mixing tank, consolidated from v3.1.

1 Purpose and scope

This document specifies how the 100 L batch mixing tank operates: equipment, parameters, master data rules, operating rules, state machine, alarms, interface variables, operator HMI, report and analytics. It is a process specification only and stands on its own.

- Valves are on/off; flow rates are constant. The scrap and recycle destinations are unlimited sinks.
- The process logic lives in the controller (Node-RED); the HMI shows state and sends commands.
- The Emergency Stop is an operator command in software and is not a safety function.
- Out of scope in this version: lot ID, level-controlled (closed-loop) dosing, keeping the product of a stopped batch (partial batches), automatic changes to order volumes.
- Every report and dashboard is headed **VC_ARG_04** (04 = fourth specification version).

2 Equipment and tags

Item	Tag	Notes
Tank	100 L	1 L = 1 % of the volume
Mixer motor	MOT	On/off
Product inlets	IN_P1, IN_P2, IN_P3	One open at a time in normal operation
Cleaning inlet	IN_CLEAN	Cleaning liquid
Final product outlet	OUT_NORMAL	Normal discharge
Scrap outlet	OUT_SCRAP	Unlimited sink
Recycle outlet	OUT_RECYCLE	Cleaning liquid to the recycling tank

3 Process parameters

Parameter	Value	Notes
Fill rate IN_P1, IN_P2	1.0 L/s	
Fill rate IN_P3	0.5 L/s	
Fill rate IN_CLEAN	1.0 L/s	To be confirmed
Discharge OUT_NORMAL	0.2 L/s	Final product
Discharge OUT_SCRAP, OUT_RECYCLE	1.0 L/s	
Production mixing time	per recipe	Integer seconds, 5–600 s
Cleaning mixing time	300 s	Configurable
Cleaning liquid volume	40 L	Fixed
Batch volume	40–100 L, integer	Minimum 40 L applies to every recipe
Fill deviation tolerance	min(5 % of the planned dose, 1.0 L)	Per product; strictly greater triggers alarm 3
Overflow limit	level > 101 L	Strictly greater
Product rounding	0.1 L, half up	Prd_1 and Prd_2 rounded; Prd_3 takes the remainder

4 Master data and order rules

Recipe attributes: Recipe ID (R###), percentages of Prd_1, Prd_2 and Prd_3 (integers adding up to exactly 100; Prd_3 may be 0), mixing time (integer seconds, 5–600) and Approved (YES or NO). Only approved recipes can run. A product with 0 % is skipped. A recipe with a missing or invalid attribute is not valid.

Planned volumes. Prd_1 and Prd_2 are the batch volume times their percentage, rounded half up to 0.1 L; Prd_3 is the batch volume minus the other two. If Prd_3 is 0 % the remainder goes to the last non-zero product.

Production order attributes: order number, Batch ID, Recipe ID, volume (integer litres). Orders run in list order.

- Batch ID is text and unique. The suffix `.n` marks the sub-batches of one order; sub-batches share the recipe and are at most 100 L each.

- Allowed batch volume 40–100 L. A pending batch with a volume outside this range, or with a recipe that is not approved or not valid, raises alarm 6 (section 7).
- A list is rejected, and no batch starts, if it has a duplicate Batch ID, an unknown recipe, a non-integer volume or a recipe with percentages not adding up to 100.

Sub-batch rebalancing rule (order preparation). When the last sub-batch of an order is below the minimum volume, the shortfall is taken from the immediately preceding sub-batch, provided that one stays at or above the minimum; otherwise from the one before it. The order total does not change. For example, 100 + 100 + 35 L becomes 100 + 95 + 40 L; 100 + 20 L becomes 80 + 40 L. A single batch below the minimum has no sibling and is handled by alarm 6. Rebalancing is done when the order list is prepared, before the first sub-batch starts; the process itself never changes an order volume.

5 Operating rules

1. **Batch start.** The HMI shows the next pending batch (number, Batch ID, recipe, planned volume, mixing time). The operator types an integer volume (40–100 L) and presses Start for each batch.
2. **Volume discrepancy.** If the typed volume differs from the order volume, the process asks for confirmation. If confirmed, the typed volume is used and the report remarks "Volume order X L, entered Y L, confirmed by operator". If cancelled, nothing starts. A typed volume out of range is rejected with no dialog.
3. **Batch allowed.** When a batch becomes the pending one (state Stopped or Waiting) it is checked: volume 40–100 L, recipe valid and approved. If not, alarm 6 is raised (rule 11). A repeat after an alarm is not checked again.
4. **Start validity.** Valid data, no active emergency alarm (1, 2 or 4), no active alarm 6, a pending batch exists, state Stopped or Waiting.
5. **Cleaning criterion.** A pre-batch cleaning is required when the recipe differs from that of the last batch actually produced, or the last recipe is unknown (after an alarm). A cancelled batch does not change the last recipe.
6. **Initial state.** Stopped, tank clean, last recipe empty. The first batch needs no cleaning.
7. **Cleaning_OP sub-steps.** (a) scrap, only if level > 0; (b1) fill IN_CLEAN to 40 L with the motor on; (b2) mix with the motor on for the cleaning mixing time; (c) drain to OUT_RECYCLE until level = 0.
8. **Normal Stop.** From Filling, Mixing or Emptying: the batch finishes, then Cleaning_OP, then Stopped. From Waiting: Cleaning_OP, then Stopped. During a pre-batch cleaning: the cleaning finishes, then Stopped, and the batch stays pending. It cannot be cancelled once pressed. An alarm cancels a pending Normal Stop.

9. **After an alarm (1 to 5) the batch is always repeated** in full (same recipe and volume, no new confirmation). An unfinished pre-batch cleaning keeps the order pending.
10. **End of list.** After the last batch the HMI shows LIST COMPLETE; Start is refused; the operator uses Normal Stop (final cleaning).
11. **Cancel batch.** With alarm 6 active the operator presses Cancel batch. The batch is closed as SKIPPED and the next order becomes pending. Start is refused while alarm 6 is active and the volume dialog cannot override it.
12. **After emergency alarms.** Alarms 1, 2 and 4 need Reset; Reset is refused while the cause is still present. Alarm 3 ends in Stopped and is acknowledged by the next Start.

6 State machine and transitions

Seven states, no Pause. Internal flags: `tankDirty`, `lastRecipe` (empty = unknown), `stopReq`, `retryPending`, `cleanReason` (PRE_BATCH, STOP, ALARM), `batchRejected` (alarm 6).

`stEtapa`	State	Meaning
0	Stopped	Clean tank, or process disabled
1	Filling_OP	Doses products one at a time (`stProducto` 1–3)
2	Mixing_OP	Motor on for the recipe's mixing time
3	Empty_OP	Normal discharge at 0.2 L/s
4	Waiting	Empty, dirty tank, between batches
5	Cleaning_OP	Sub-step in `stSubLimpieza` (0 none, 1 scrap, 2 fill, 3 mix, 4 drain)
9	Alarm Stop	Emergency alarms 1, 2 and 4

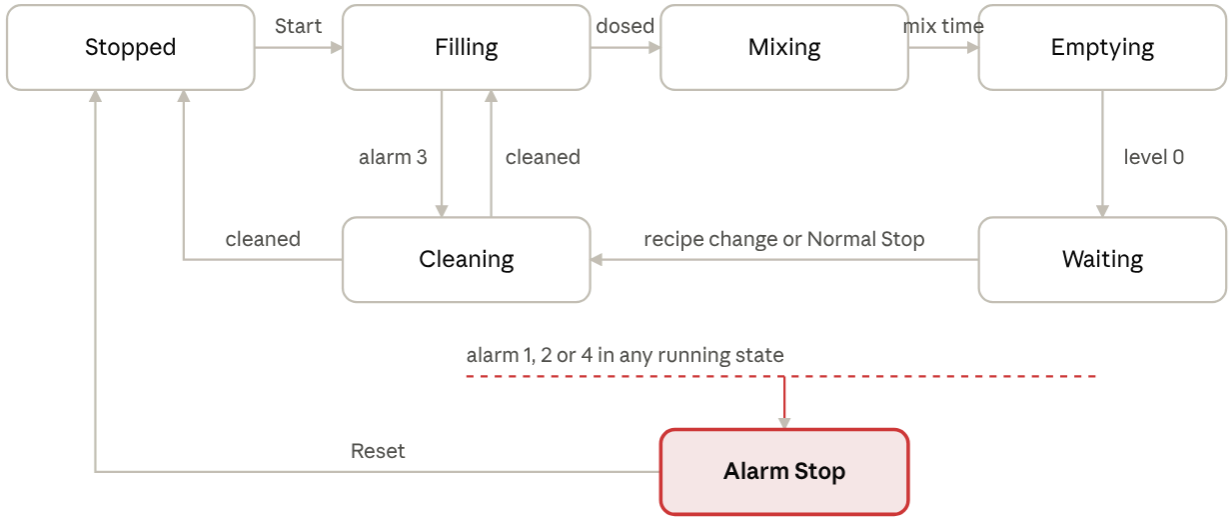
#	From	Event / condition	To	Actions
T1	Stopped	Start valid, tank clean	Filling_OP	Row IN stamped
T2	Stopped	Start valid, tank dirty (after an emergency alarm)	Cleaning_OP (PRE_BATCH)	
T3	Waiting	Start valid, same recipe as last	Filling_OP	

#	From	Event / condition	To	Actions
T4	Waiting	Start valid, different recipe	Cleaning_OP (PRE_BATCH)	
T5	Filling_OP	Product dosed within tolerance, more products follow	Filling_OP (next product)	Skips 0 % products
T6	Filling_OP	Last product dosed within tolerance	Mixing_OP	
T7	Mixing_OP	Recipe mixing time elapsed	Empty_OP	
T8	Empty_OP	Level 0, no Normal Stop pending	Waiting	Row COMPLETED, OUT stamped, tank dirty, last recipe set
T9	Empty_OP	Level 0, Normal Stop pending	Cleaning_OP (STOP)	As T8, then cleaning
T10	Waiting	Normal Stop	Cleaning_OP (STOP)	
T11	Filling, Mixing, Empty	Normal Stop	same	`stopReq` set
T12	Cleaning_OP (PRE_BATCH)	Normal Stop	same	`stopReq` set
T13	Cleaning_OP (PRE_BATCH)	Finished, no `stopReq`	Filling_OP	Tank clean, cleaning time to the row
T14	Cleaning_OP	Finished with `stopReq`, or reason STOP or ALARM	Stopped	Tank clean
T15	Filling_OP	Alarm 3	Cleaning_OP (ALARM)	Row ABORTED, scrap, clean, Stopped, repeat pending
T16	Any running state	Alarm 1, 2 or 4	Alarm Stop	Everything off, level frozen; row ABORTED if a batch was active; alarm 5 if product was already discharged

#	From	Event / condition	To	Actions
T17	Alarm Stop	Reset, cause cleared	Stopped	Tank dirty, last recipe unknown
T18	Stopped (after alarm 3)	Start	Filling_OP	Acknowledges alarm 3; repeat as a new row
T19	Stopped, Waiting	Pending batch not allowed (volume or recipe)	same	Alarm 6 raised; Start refused
T20	Stopped, Waiting	Cancel batch with alarm 6 active	same	Row SKIPPED, alarm cleared, next order pending and checked

"Any running state" means Filling, Mixing, Emptying, Waiting and Cleaning (not Stopped).

A batch runs Filling, Mixing, Emptying; alarms divert it to cleaning or Alarm Stop



state machine · 7 states, main transitions

Waiting also goes straight to Filling (T3) when the next recipe equals the last one. Alarm 3 diverts a batch to a cleaning; alarms 1, 2 and 4 stop the machine until Reset. Alarm 6 (T19, T20) changes no state: the machine waits in Stopped or Waiting for Cancel batch.

7 Alarms 1 to 6

Code	Alarm	Class	Detection
0	No alarm		
1	Overflow	Emergency (Alarm Stop)	Level crosses 101 L with any inlet open; latched until Reset. Also active during cleaning
2	Emergency Stop	Emergency (Alarm Stop)	Operator command in any running state, including Waiting
3	Fill deviation	Controlled	After each product, |delivered − planned| > min(5 % of planned, 1.0 L). Delivered = level after − level before
4	Step timeout / no progress	Emergency (Alarm Stop)	See below
5	Partial batch	Warning, raised with its cause	Alarm 1, 2 or 4 during Emptying after product was discharged to OUT_NORMAL
6	Batch not allowed	Warning, blocks Start (no Alarm Stop)	The pending batch has a volume below 40 L or above 100 L, or a recipe that is not approved or not valid. Raised when the batch becomes pending; cleared by Cancel batch

Alarm 3. The batch is aborted, its content goes to scrap, the tank is cleaned and the machine stops; the next Start repeats the batch.

Alarm 4 (progress monitor). Active in Filling (during a dose), Emptying and the cleaning scrap, fill and drain steps. Over a 15 s window the level must move in the expected direction by at least 20 % of the expected change (expected rate = nominal flow of the open valve). A step also times out when it exceeds 1.5 × its expected duration + 10 s. It detects an inlet with no flow, an outlet stuck closed, and the deadlock where a stuck-open inlet (1 L/s) equals the scrap or recycle drain (1 L/s). Mixing is time-controlled and not monitored. Reset needs the cause cleared.

Alarm 6. The alarm text names the reason: `Volume 35 L below minimum 40 L`, `Volume 120 L above maximum 100 L`, `Recipe R### not approved`, `Recipe R### not valid`. The machine stays in Stopped or Waiting with the tank unchanged; Normal Stop and

Emergency Stop (from Waiting) keep working. The only way forward is Cancel batch. The batch is not repeated and does not count as aborted.

Priority. 1, 2 and 4 before 3. Alarm 5 is always raised together with its cause. Alarm 6 can only occur when no batch is running.

Overflow precedence. On a full 100 L batch the last product has only 1 L of headroom below the limit, so an excess flow on it raises alarm 1 before the end-of-dose check can raise alarm 3.

Reset and Start. Reset is refused while the cause of an emergency alarm is still present. After alarm 3, Start is accepted even if the cause is still present; the repeat then trips again.

8 Output matrix

State / sub-step	IN_P1	IN_P2	IN_P3	IN_CLEAN	OUT_NORMAL	OUT_SCRAP	OUT_RECYCLE	MOT
Stopped, Waiting, Alarm Stop	0	0	0	0	0	0	0	0
Filling_OP	one of the three, matching `stProducto`			0	0	0	0	0
Mixing_OP	0	0	0	0	0	0	0	1
Empty_OP	0	0	0	0	1	0	0	0
Cleaning (a) scrap	0	0	0	0	0	1	0	0
Cleaning (b1) fill	0	0	0	1	0	0	0	1
Cleaning (b2) mix	0	0	0	0	0	0	0	1
Cleaning (c) drain	0	0	0	0	0	0	1	0

Alarm 6 changes no output.

9 Scrap, partial batch and cancelled batch

Scrap is product drained through OUT_SCRAP (the scrap step of a cleaning). Cleaning liquid sent to OUT_RECYCLE is not scrap. Scrap is attributed to the row that caused it: the scrap of a pre-batch cleaning to the batch about to start, the scrap of a post-alarm

cleaning to the aborted row. The report shows it in Remarks ("Scrap x.x L") and its summary; the HMI and the analytics show the scrap of the current batch and the running total.

Partial batch. When alarm 1, 2 or 4 stops a batch during Emptying after product has gone to OUT_NORMAL, the row is ABORTED and ALARMS holds the cause and `5 – Partial batch`. Remarks gives the litres already discharged and left in the tank. After Reset the next Start cleans (the left-over product goes to scrap) and repeats the batch in full.

Cancelled batch (alarm 6). The row keeps STATUS `SKIPPED` and ALARMS `6 – Batch not allowed`. Remarks gives the reason, the ordered volume and "Cancelled by operator". IN, OUT, process time, cleaning time and produced volume stay blank. There is no repeat row. The ordered volume counts as **not produced**. The last recipe and the tank state are unchanged, so the next batch is compared with the last batch actually produced.

10 Interface variables

Message contract between controller and HMI: readings `topic: estado` (full snapshot), commands `topic: orden`.

Commands (HMI to controller)

Variable	Type	Notes
cmdMarcha	BOOL pulse	Start, carries `volLote_L`
volLote_L	UINT	40–100, integer
cmdConfirmaVol / cmdCancelaVol	BOOL pulse	Answer to the volume dialog
cmdParo	BOOL pulse	Normal Stop
cmdParoEmerg	BOOL pulse	Emergency Stop
cmdRearme	BOOL pulse	Reset
cmdCancelaLote	BOOL pulse	Cancel batch; accepted only with alarm 6 active

Readings (controller to HMI)

Variable	Type	Notes
pvNivel_L	REAL	Normal range 0 to about 103 L; the HMI accepts 0–130
stEtapa	UINT	0–5, 9
stSubLimpieza, stProducto	UINT	0–4 and 0–3
stMotivoLimpieza	UINT	0 none, 1 pre-batch, 2 stop, 3 after alarm
stIN_P1, stIN_P2, stIN_P3, stIN_CLEAN	BOOL	
stOUT_NORMAL, stOUT_SCRAP, stOUT_RECYCLE	BOOL	
stAgitador	BOOL	
stAlarma	UINT	0–6
stAlarmaOrigen	UINT	0 none, 1 machine, 2 operator
stAlarmaTexto	STRING	Detail, including the alarm 6 reason
stMensaje	STRING	Process feedback (rejections, refusals)
stTanqueSucio, stParoSolicitado, stReintento	BOOL	
stConfirmaVol, stListaCompleta	BOOL	Volume dialog; no pending batch
stLoteParcial	BOOL	Alarm 5 active
stLoteRechazado	BOOL	Alarm 6 active
recetaAprobada	BOOL	
loteIdx, loteTotal	UINT	
loteId, recetaId, ultimaReceta	STRING	
volOrden_L, volLote_L	REAL	Order volume, entered volume
recPct_P1..P3	REAL	Recipe percentages of the current batch
tMezclaRec_s	REAL	Mixing time of the current recipe
plan_P1..P3_L, dos_P1..P3_L	REAL	Planned and delivered

Variable	Type	Notes
volDescargado_L	REAL	Litres discharged to OUT_NORMAL for the current or last batch
volScrapLote_L, volScrapTotal_L	REAL	Scrap of the current batch, running total
tEtapa_s	REAL	Elapsed time in the step
kpiLotesOK, kpiLotesAbort, kpiLotesCancel, kpiLimpiezas	UINT	Counters
kpiUltimoCiclo_s, volNoProducido_L	REAL	Last cycle time; litres not produced

The HMI rejects an incomplete or out-of-range snapshot, names the first offending field and keeps the last valid reading.

11 Operator HMI requirements

- **Process view:** tank with a 0–100 L scale and a batch-volume marker (a 40 L marker during cleaning), four inlet valves, three outlet valves, motor and agitator, flow animation. Every state also appears as text (OPEN, CLOSED, RUNNING, STOPPED). Cleaning liquid is shown in a distinct colour.
- **Batch data:** Batch ID, recipe, order volume, mixing time, planned and delivered volume per product, litres discharged, scrap of the current batch and running total.
- **Alarm field:** code, name and detail text for alarms 1 to 6; Alarm 5 and 6 are shown with the alarm that caused them or with the batch that was rejected.
- **Volume entry:** integer field, 40–100 L, validated before Start; the confirmation dialog of rule 2.
- **Buttons and when they are enabled:**

Button	Enabled
Start	Stopped or Waiting, allowed batch, valid volume (a repeat keeps the interrupted volume)
Normal Stop	Filling, Mixing, Emptying, Waiting, pre-batch cleaning; once per batch
Emergency Stop	Every running state
Reset	Alarm Stop only
Cancel batch	Alarm 6 only

- Each valid press sends one `true` pulse with a 500 ms guard against double sends; nothing is sent on release.
- Footer text: "Emergency Stop is not a safety function".

12 Production Order Report

One record per production order, created as PENDING when the list is loaded and filled as production progresses. Header: VC_ARG_04.

Column	Content
`#`	Record number. Orders are 1 to N; repeats are appended at the end
Batch ID	Text, for example `0006.1`
Recipe	Recipe ID
Vol. order (L)	Volume of the order
Vol. produced (L)	Sum of the delivered doses of a COMPLETED batch; blank otherwise
IN	Start of Filling_OP, `dd.mm.yyyy hh:mm:ss`
OUT	End of Empty_OP (COMPLETED) or the alarm instant (ABORTED)
Process time (s)	OUT − IN (fill, mix, empty only)
Cleaning time (s)	Completed cleaning associated with the record (see below)

Column	Content
STATUS	`PENDING`, `RUNNING`, `COMPLETED`, `ABORTED (ALARM)`, `SKIPPED` (cancelled after alarm 6)
ALARMS	`code – name`, several separated by `;`, for example `2 – Emergency Stop; 5 – Partial batch` or `6 – Batch not allowed`
Remarks	Alarm detail, volume discrepancy, retry reference, scrap, partial discharge, cancellation reason, final cleaning

Rules:

- An aborted batch keeps its record as ABORTED. The repeat is a new record appended at the end with the same Batch ID and Remarks `Retry of #m`, where m is the record it repeats. Repeats are not in chronological order; the timestamps are.
- A cancelled batch (alarm 6) keeps its record with STATUS `SKIPPED`, ALARMS `6 – Batch not allowed` and the reason and "Cancelled by operator" in Remarks. IN, OUT, process time, cleaning time and produced volume are blank; there is no repeat record.
- Cleaning time is the pre-batch cleaning completed before the batch started. On an aborted record it also includes the cleaning that follows its alarm; when both exist Remarks gives the breakdown. A cleaning after a Normal Stop is written in the Remarks of the last batch. A cleaning interrupted by an alarm is noted in the Remarks of the next record.
- A fill deviation inside the tolerance is written in Remarks. A produced volume different from the order volume gets a Remarks comment.
- Scrap is written as `Scrap x.x L` in the Remarks of the record that caused it.
- A summary accompanies the report: records, completed, aborted, skipped, alarms by code 1 to 6, litres produced, litres not produced, total cleaning time, first IN, last OUT, duration, cleanings completed, scrap generated, partial batches.
- An event log keeps every command, state change, alarm and refusal with its clock time, including alarms that have no record (for example during a pre-batch cleaning).

13 Production analytics

A dashboard headed **VC_ARG_04** presents the report and the event log. Definitions:

Indicator	Definition
Batches completed, aborted, skipped	Records by STATUS

Indicator	Definition
Alarms raised	Count per code 1 to 6 from the event log
Cleanings finished	Cleanings that reached the end of the drain step
Litres produced	Sum of Vol. produced
Litres not produced	Sum of Vol. order of SKIPPED records
Scrap generated	Litres drained through OUT_SCRAP
Time making good product	Share of the elapsed time spent in Filling, Mixing and Emptying of batches that were completed

Views: tank level over time with the overflow limit, a state strip and alarm markers by code; time by activity; alarms by type; valve and motor on-time; litres per batch (produced, ordered and aborted, cancelled shown distinctly); cumulative litres produced and scrap; alarm log; report table. A legend is shown whenever there are two or more series, status is never by colour alone, and every chart has a tooltip.

14 Open points

To confirm

1. Fill rate of IN_CLEAN (1.0 L/s).
2. Cleaning mixing time (300 s, configurable).
3. Mixing time range 5–600 s.
4. The 40 L minimum for every recipe, or a minimum per recipe.
5. Alarm 6 also covers volume above 100 L and recipes that are not approved or not valid; the volume dialog cannot override it.
6. The header number follows the specification number (VC_ARG_04).

Future improvements

- Partial batches: keep the product and resume.
- Mixing time depending on the batch volume (today one value per recipe).
- Merging a cancelled volume into the next order of the same recipe; automatic rebalancing of sub-batches.
- Level-controlled (closed-loop) dosing; lot ID.
- A report record for an alarm that occurs during a pre-batch cleaning (today only in the event log).